

Predictors of Opioid Misuse Disorder in the US

Final Project

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Note

From Alex: Some sections have our names next to the section title and others don't. I'm fine with either, but could we make them consistent?

Naomi: We should remove our names after we're done.

1 Abstract

Note

Naomi to complete. Use 250 words or less.

Background: Opioid addiction is a public health crisis that has affected countless lives.

Research Question: What factors are important for predicting prevalence rate of opioid use disorder (OUD)?

Methods: We downloaded data on the prevalence of pain reliever misuse in each state from the National Survey on Drug Use and Health from the Substance and Mental Health Services Administration for all available years (2018–2019 and 2021–2024). We performed exploratory data analysis to understand the distribution of our outcome variable (prevalence of pain reliever misuse) and the relationships between our predictors and the outcome. We used Box-Cox transformations to normalize data distributions of numeric predictors, and then split the data 70:30 into training and test sets. We built three regression models to predict the prevalence of pain reliever misuse: a full multiple linear regression model with all predictors, a stepwise selection model based on AIC, and a LASSO regression model with hyperparameter tuning...

Results: [TBD]

Conclusion: [TBD]

2 Key words

Opioid Use Disorder, Regression Analysis, Public Health, Substance Use Disorder

3 Introduction

Opioid addiction is a public health crisis that has affected countless lives. The United States is experiencing a public health crisis of almost unprecedented scale: an epidemic of opioid use disorder (OUD) and related overdose deaths, the magnitude of which has increased exponentially in the past decades (Mancher & Leshner, 2019). In just one year, the age-adjusted rate of drug overdose deaths increased 31% (from 21.6 in 2019 per 100k population to 28.3 in 2020) (Hedegaard et al., 2021). In this project, we investigate what factors are important for predicting the prevalence rate of OUD across US states.

4 Literature review

OUD is a complex problem with many risk factors, including sociodemographic, geographic, and community/societal factors (Jalali et al., 2020). Sociodemographic risk factors include age, sex, and occupation. Opioid misuse is most common in young adults (18-25 years old), and first misuse of opioids at an early age is a significant risk factor for OUD (Crane, 2013; Schepis & Hakes, 2017). In addition, male sex is a risk factor for OUD because men are more likely than women to experience euphoria from opioids (Comer et al., 2010). Some occupations, particularly those that involve heavy physical labor (e.g., construction, mining), are also risk factors for OUD, because they may be associated with more work-related injuries that lead to chronic pain and treatment with opioid analgesics (Friedman et al., 2025; Shaw et al., 2020). In addition to these sociodemographic factors, mental health conditions, such as anxiety and depression, and a history of misusing other substances (eg, illicit drugs) can increase the risk of OUD (Cicero et al., 2020; Valente et al., 2020).

The risk of OUD also varies by geographic location. For example, opioid medications are prescribed more frequently in non-metropolitan areas, which may be due to the higher proportion of older individuals in these areas who need pain relief after surgeries or individuals who are injured while performing manual labor [Keyes et al. (2014)](Rigg & Monnat, 2015). Rural areas may also have fewer community resources for prevention and treatment programs than urban areas, which can increase the likelihood of developing and prolonging OUD (*Preventing Prescription Drug Misuse*, 2016).

Furthermore, disparities in healthcare access and insurance coverage can make it more difficult for some individuals to receive care for OUD, particularly those who are racial or ethnic minorities, live in rural areas, or cannot afford health insurance *Opioid Use Disorder* (2019). The US Affordable Care Act (ACA; also known as “Obamacare”) provided states the option to extend Medicaid benefits to non-elderly adults with lower incomes, including access to OUD treatment; however, not all

states have implemented the expansion (Mitchell, 2021). These differences in the implementation of ACA have resulted in a lower rate of prescriptions for medications to treat OUD in states that have not yet expanded Medicaid compared with those that have (Lewis et al., 2020; Sharp et al., 2018).

Although many studies have documented risk factors for OUD in various groups (e.g., military veterans) or geographic regions (e.g, urban versus rural), few have analyzed nationwide data. One study, which modeled National Survey on Drug Use and Health (NSDUH) data from 2017-2018 using multivariate logistic regression, found that the most significant predictors of OUD were abuse of other substances, particularly tranquilizers (odds ratio [OR]=16.7), criminal history (OR=2.6), and lack of private health insurance (OR=2.1) (all $P < 0.001$) (Montiel Ishino et al., 2020). Here, we perform a similar analysis using a broader range of NSDUH data (2016-2019 + 2022), including data before and after the Covid pandemic.

5 Methods

Note

Discuss the key aspects of your problem, data set and regression model(s). Given that you are working on real-world data, explain at a high-level your exploratory data analysis, how you prepared the data for regression modeling, your process for building regression models, and your model selection.

We use data on the state-level prevalence of pain reliever misuse and demographic characteristics from the National Survey on Drug Use and Health (NSDUH) from the Substance and Mental Health Services Administration for 2018-2019 and 2021-2024 (Substance Abuse and Mental Health Services Administration (SAMHSA)'s Data Analysis System (DAS), 2018-2024).

5.1 Exploratory Data Analysis

We performed exploratory data analysis to understand the distribution of our outcome variable (prevalence of pain reliever misuse) and the relationships between our predictors and the outcome. We visualized the correlations between predictors and the outcome variable, as well as the distribution of the numeric variables.

5.2 Data Preparation

Box-Cox transformations were used to normalize data distributions of numeric predictors. We confirmed that these variables had positive values, which is required for this method. We explicitly calculated the optimal lambda values, which provided insight into the optimal transformations needed to normalize each variable, and then used these values to calculate the transformed data.

The data did not contain any missing values, so imputation was not needed. For consistency with a

similar study (Montiel Ishino et al., 2020) (see Literature review section), which will be used for comparison in the Discussion section, no additional features were created.

Finally, we split the data 70:30 into training and test sets.

5.3 Assumptions (Nana)

5.4 Model Building

For our models, we will attempt to predict the state-level prevalence of pain reliever misuse by evaluating a series of regression techniques designed to balance model fit with predictive accuracy. Our methodology follows a multi-stage approach, transitioning from a comprehensive baseline to restricted models that handle the complexities of high-dimensional demographic data.

5.4.1 Baseline Full Multiple Linear Regression

Our initial approach involves constructing a Full Multiple Linear Regression model using all 20+ available predictors. This process partitions the response variable into a systematic component represented by the predictors and a random error component. We use the method of Least Squares to calculate the unique set of coefficients that minimizes the sum of squared residuals, ensuring we obtain the best linear unbiased estimates (BLUE) given our data. To execute this, we will fit a model using every numeric demographic and socioeconomic variable in the training set to establish a performance ceiling for our R^2 and residual standard error.

5.4.2 Stepwise Variable Selection

To address potential redundancy in our 20+ predictors, we implement Stepwise Selection based on the Akaike Information Criterion (AIC). This “auto-selection” method iteratively adds or removes variables, comparing a larger model (Ω) to a smaller subset (ω). The goal is to minimize the AIC, which effectively penalizes the model for every additional parameter included, thereby favoring simpler, more interpretable explanations.

5.4.3 LASSO Regression

Because demographic variables (e.g., age, race, and poverty levels) are often highly correlated, we will utilize LASSO (Least Absolute Shrinkage and Selection Operator). This technique introduces a penalty term to the OLS equation that can shrink insignificant coefficients exactly to zero. This process requires Hyperparameter Tuning to find the optimal value of λ (the penalty strength) that produces the minimum error.

5.4.4 Predictive Validation and Testing

To ensure our models generalize to unseen data, we utilize a two-stage validation process. First, we will implement 10-fold Cross-Validation on the training data. This involves splitting the training set

into ten equal parts, or “folds”; the model is trained on nine folds and validated on the tenth, with the process repeating ten times so every data point is used for validation once. We use the average Root Mean Squared Error (RMSE) across these folds to select our winning model. Finally, we will test the winning model on the held-out test set to get an honest estimate of its predictive ability on new data.

5.5 Model Selection (Said)

6 Results

Note

Describe the specifics of what you did (data exploration, data preparation, model building, model selection, model evaluation, etc.), and what you found out (statistical analyses, interpretation and discussion of the results, etc.).

```
# Read in all CSVs from the data directory and store them in a list
filepaths <- list.files(path = "data", pattern = "*.csv", full.names = TRUE)

# Helper to extract the "YY-YY" year substring from filenames
extract_year_tag <- function(path) {
  fname <- basename(path)
  tag <- stringr::str_extract(fname, "\\d{2}-\\d{2}")
  if (is.na(tag)) tag <- gsub("\\.csv$", "", fname)
  tag
}

# Read each file and add a 'year' column based on the filename (e.g. "18-19")
data_list <- lapply(filepaths, function(fp) {
  df <- readr::read_csv(fp, show_col_types = FALSE) |>
    janitor::clean_names()

  # Get name of first column (variable of interest)
  first_col <- names(df)[1]

  # Transform each df.
  df <- df |>
    # Filter out redundant ("Overall") rows
    filter(
      state_us_abbreviation != "Overall") |>
```

```
# Calculate precise percentages
group_by(state_us_abbreviation) |>
mutate(
  denominator = weighted_count[column_percent == 1]
) |>
ungroup() |>
mutate(
  percent = weighted_count / denominator
) |>

# Get subset of columns
select(1, state_us_abbreviation, percent) |>

# Pivot first column wider
mutate(
  "{first_col}" := paste0(first_col, "_", as.character(.data[[first_col]]))
) |>
pivot_wider(names_from = 1, values_from = percent) |>

# Drop columns with all the same values
select(where(~ n_distinct(.) > 1)) |>

# Add year column
mutate(
  year = extract_year_tag(fp)
) |>

# Clean names
janitor::clean_names()

df
})

# Get list of data years
years <- sapply(filepaths, extract_year_tag) |> unique()

# Iterate across years, join data frames in data_list by state and year within each year.
yearly_joined_data <- lapply(years, function(yr) {
  year_dfs <- data_list |>
```

```

    purrr::keep(~ n_distinct(.x$year) == 1 && .x$year[[1]] == yr)

if (length(year_dfs) == 0) return(NULL)
if (length(year_dfs) == 1) return(year_dfs[[1]])

Reduce(function(x, y) full_join(x, y, by = c("state_us_abbreviation", "year")), year_dfs)
})

# Append yearly_joined_data across all years into a single data frame
# Filter out NULL values first to avoid bind_rows errors
df <- yearly_joined_data |>
  purrr::discard(is.null) |>
  bind_rows()

df_renamed <- df |>
  rename(
    state = state_us_abbreviation,

    age_12_17 = rc_adult_age_category_recode_5_12_17_years_old,
    age_18_25 = rc_adult_age_category_recode_1_18_25_years_old,
    age_26_34 = rc_adult_age_category_recode_2_26_34_years_old,
    age_35_49 = rc_adult_age_category_recode_3_35_49_years_old,
    age_50_plus = rc_adult_age_category_recode_4_50_or_older,

    medicaid_coverage = covered_by_medicaid_chip_1_yes,

    less_than_high_school = rc_education_categories_1_less_high_school,
    high_school = rc_education_categories_2_high_school_grad,
    some_college = rc_education_categories_3_some_coll_assoc_dg,
    college_grad = rc_education_categories_4_college_graduate,

    emp_full_time = employment_status_imputation_revised_1_employed_full_time,
    emp_part_time = employment_status_imputation_revised_2_employed_part_time,
    emp_unemployed = employment_status_imputation_revised_3_unemployed,

    pain_reliefers_misuse = rc_pain_reliefers_past_year_misuse_1_misused_within_the_past_year,

    race_nh_white = rc_race_recode_4_levels_1_white_not_hispanic,
    race_nh_black = rc_race_recode_4_levels_2_black_not_hispanic,

```

```

race_hisp = rc_race_recode_4_levels_4_hispanic,
race_other = rc_race_recode_4_levels_3_other_or_multiple_not_hispanic,

poverty = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_1_living_in_pover
poverty_2x = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_2_income_up_to
poverty_2x_plus = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_3_income
) |>

select(
  state,
  year,
  pain_relievers_misuse,
  age_12_17, age_18_25, age_26_34, age_35_49, age_50_plus,
  medicaid_coverage,
  less_than_high_school, high_school, some_college, college_grad,
  emp_full_time, emp_part_time, emp_unemployed,
  race_nh_white, race_nh_black, race_hisp, race_other,
  poverty, poverty_2x, poverty_2x_plus
)

```

6.1 Exploratory Data Analysis

The data set contains 204 state-year rows and 23 features, and is 100 percent complete. The data are all numeric, and the target variable is `pain_relievers_misuse`, which represents the percentage of individuals who misused pain relievers in the past year. The other variables represent various demographic and socioeconomic characteristics of people surveyed in the state and year. See Table 1 for descriptive characteristics of the data.

```

tbl_summ_stats <- df_renamed |>
  skim_without_charts() |>
  janitor::clean_names() |>
  as_tibble() |>
  rename(Variable = skim_variable,
    "Completeness rate" = complete_rate,
    "Mean" = numeric_mean,
    "SD" = numeric_sd,
    "P50" = numeric_p50,
    "Min" = numeric_p0,
    "P25" = numeric_p25,
    "P75" = numeric_p75,

```



```

    "Max" = numeric_p100) |>
  select(Variable, "Completeness rate", Mean, Min, P25, P50, P75, Max) |>
  mutate(across(where(is.numeric), ~ signif(., 2))) |>
  knitr::kable()

tbl_summ_stats

```

Table 1: Summary statistics for all variables in the dataset

Variable	Completeness rate	Mean	Min	P25	P50	P75	Max
state	1	NA	NA	NA	NA	NA	NA
year	1	NA	NA	NA	NA	NA	NA
pain_relievers_misuse	1	0.033	0.0120	0.026	0.032	0.039	0.058
age_12_17	1	0.091	0.0540	0.085	0.091	0.096	0.120
age_18_25	1	0.120	0.0980	0.120	0.120	0.130	0.160
age_26_34	1	0.140	0.1200	0.130	0.140	0.140	0.250
age_35_49	1	0.220	0.2000	0.210	0.220	0.230	0.260
age_50_plus	1	0.430	0.3100	0.410	0.430	0.440	0.500
medicaid_coverage	1	0.190	0.0500	0.150	0.190	0.230	0.370
less_than_high_school	1	0.084	0.0460	0.068	0.079	0.097	0.160
high_school	1	0.250	0.1300	0.220	0.250	0.270	0.370
some_college	1	0.280	0.1500	0.270	0.280	0.300	0.380
college_grad	1	0.290	0.1500	0.260	0.290	0.330	0.590
emp_full_time	1	0.440	0.2700	0.420	0.440	0.460	0.570
emp_part_time	1	0.130	0.0790	0.120	0.130	0.140	0.190
emp_unemployed	1	0.042	0.0170	0.033	0.041	0.050	0.075
race_nh_white	1	0.680	0.2100	0.580	0.710	0.800	0.940
race_nh_black	1	0.110	0.0042	0.032	0.072	0.150	0.440
race_hisp	1	0.120	0.0150	0.051	0.092	0.140	0.480
race_other	1	0.089	0.0230	0.047	0.065	0.095	0.680
poverty	1	0.150	0.0710	0.110	0.140	0.170	0.270
poverty_2x	1	0.190	0.0900	0.170	0.190	0.220	0.270
poverty_2x_plus	1	0.660	0.5000	0.620	0.670	0.720	0.780

```

# Get correlation matrix
df_renamed_cor <- df_renamed |>
  select(-state, -year) |>
  cor(use = "pairwise.complete.obs")

```

```
# Suppress small correlations by setting them to NA so corrplot leaves them blank
df_renamed_cor[abs(df_renamed_cor) < 0.05] <- NA

corrplot(df_renamed_cor,
  method = "circle",
  type = "upper",
  tl.cex = 0.4,
  insig = "blank",
  na.label = " ",
  addCoef.col = "black",
  number.cex = 0.3,
  tl.srt = 45)

most_corr_w_target <- df_renamed_cor |>
  as.data.frame() |>
  rownames_to_column(var = "Variable") |>
  filter(Variable != "pain_relievers_misuse") |>
  transmute(
    Variable,
    Correlation = pain_relievers_misuse,
    Abs_Correlation = abs(pain_relievers_misuse)
  ) |>
  arrange(desc(Abs_Correlation)) |>
  filter(Abs_Correlation > 0.05) |>
  pull(Variable)
```

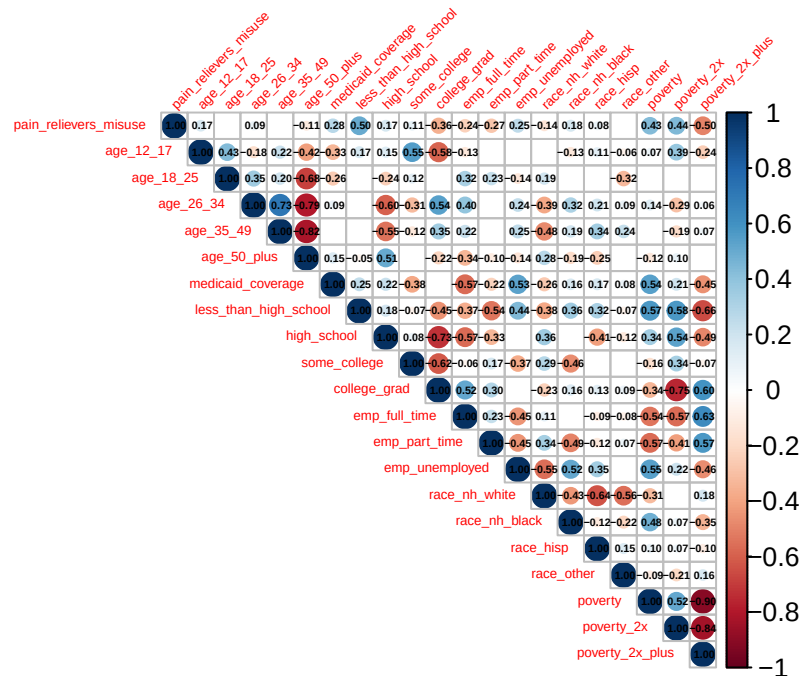


Figure 1: Correlation matrix of all non-ID variables in the dataset. Correlations < 0.05 are suppressed.

The correlation between all variables are shown in Figure 1. The following predictors are most correlated with the target variable: less_than_high_school, poverty_2x_plus, poverty_2x, poverty, college_grad, medicaid_coverage, emp_part_time, emp_unemployed, emp_full_time, race_nh_black, high_school, age_12_17, race_nh_white, age_50_plus, some_college, age_26_34, race_hisp, which may be predictors of pain reliever misuse.

Lastly, we also looked at the distribution of each variable to check for skewness and outliers.

```
# Keep only continuous
continuous_vars <- df_renamed |>
  select(where(is.numeric)) |>
  names()

df_renamed |>
  select(all_of(continuous_vars)) |>
  pivot_longer(cols = everything(), names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ variable, scales = "free", ncol = 4) +
  theme_minimal() +
  theme(panel.grid = element_blank()) +
```

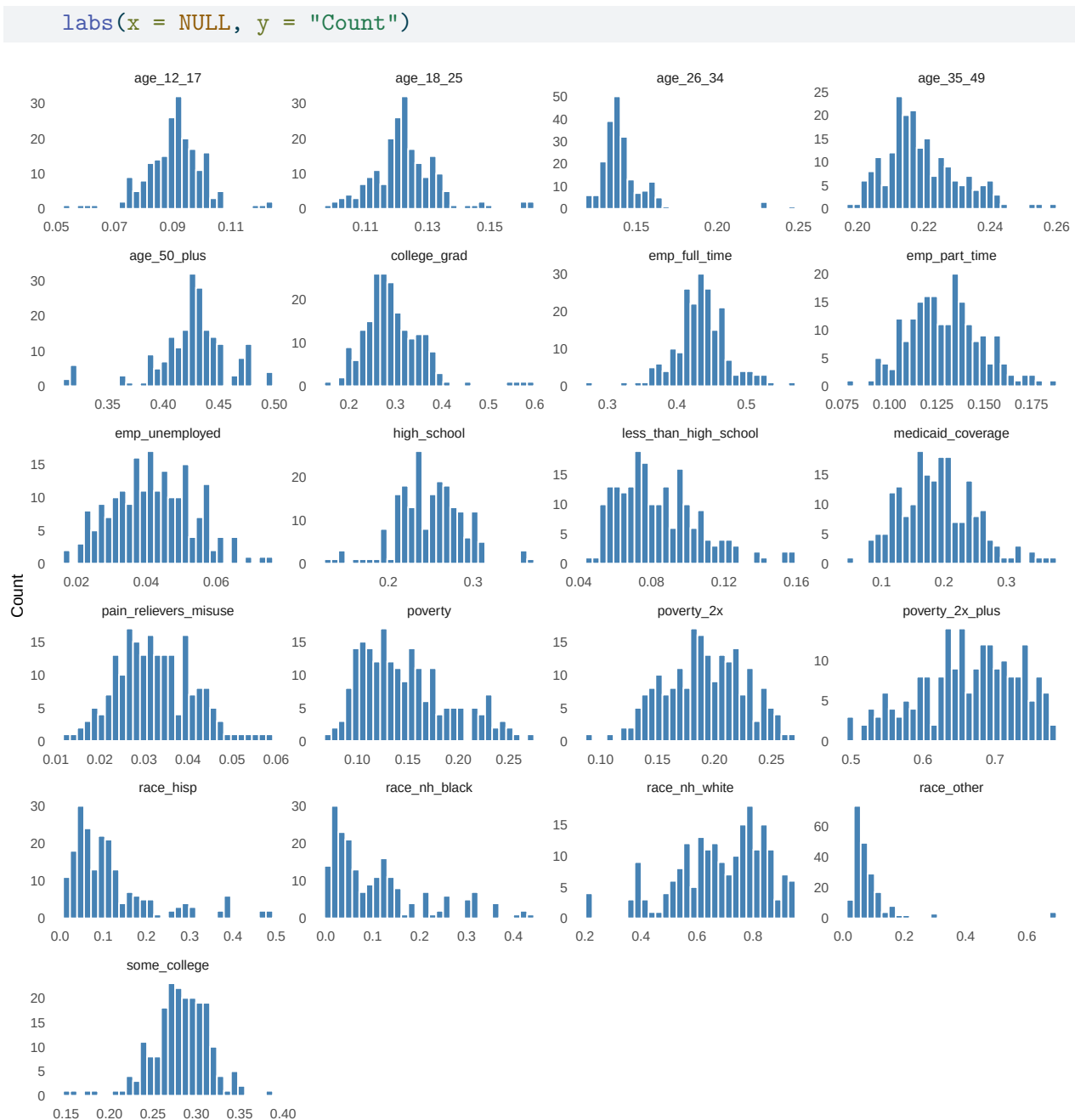


Figure 2: Distribution of numeric variables in the dataset.

Most variables are approximately normally distributed, except for many of the race indicator variables. The proportion of individuals who are non-white—especially those who are an “other” race besides hispanic, non-hispanic white, or non-hispanic black—are skewed right, mostly close to 0 across states and years, with a few higher outliers. This is expected given the racial demographics of the US. The proportion of individuals who are aged 26-34 also has a few high outliers.

6.2 Results of Data Preparation

6.2.1 Transformations

The optimal lambda values for Box-Cox transformation provided insight into which variables would benefit from various types of transformation:

- `age_50_plus`, `some_college`, `poverty_2x_plus`, and `race_nh_white` had $\lambda \approx 2$, which indicates squared transformation,
- `emp_unemployed` and `medicaid_coverage` had $\lambda \approx 0.5$, which indicates square root transformation,
- `race_nh_black`, `race_hisp`, and `poverty` had $\lambda \approx 0$, which indicates natural log transformation,
- `less_than_high_school`, `college_grad`, and `race_other` had $\lambda \approx -0.5$, which indicates reciprocal square root transformation,
- `age_18_25` had $\lambda \approx -1$, which indicates inverse transformation, and
- `age_26_34` and `age_35_49` had $\lambda = -2$, which indicates inverse squared transformation

Several variables (e.g., `age_12_17`, `poverty_2x`, `high_school`, `emp_full_time`) had $\lambda \approx 1$, which indicates that their distributions are approximately normal, so no (or very little) transformation is needed.

```
# Numeric predictors
predictors_untransformed <- select(df_renamed, age_12_17:poverty_2x_plus)

# Determine optimal lambda values for Box-Cox transformation
lambdas <- purrr::map_dbl(predictors_untransformed, ~ {
  caret::BoxCoxTrans(.x)$lambda
})

# Convert results vector to dataframe and format
optimal_lambdas <- as.data.frame(lambdas)
optimal_lambdas$variable <- rownames(optimal_lambdas)
rownames(optimal_lambdas) <- NULL
optimal_lambdas |>
  arrange(desc(lambdas)) |>
  select(variable, 'lambda' = 'lambdas') |>
  knitr::kable()
```

variable	lambda
age_50_plus	2.0
some_college	2.0
poverty_2x_plus	2.0
race_nh_white	1.9
age_12_17	1.4
emp_full_time	1.3
high_school	1.2
poverty_2x	1.2
emp_unemployed	0.7
medicaid_coverage	0.5
emp_part_time	0.3
race_nh_black	0.2
race_hisp	0.0
poverty	-0.2
less_than_high_school	-0.4
college_grad	-0.5
race_other	-0.6
age_18_25	-1.1
age_26_34	-2.0
age_35_49	-2.0

These lambda values were then used to apply Box-Cox transformation to the numeric predictor variables.

```
# Box-Cox transformation of numeric variables using optimal lambda values (previous block)
lambdas_vec <- optimal_lambdas$lambda
predictors_transformed <- purrr::map2_df(
  .x = predictors_untransformed,
  .y = lambdas_vec,
  .f = ~forecast::BoxCox(.x, .y)
)

# Restore the categorical and target variables and create new dataframe
df_prepared <- bind_cols(df_renamed[, 1:3], predictors_transformed)

# Removing the year and state variables since they are not needed for modeling and may cause d
df_prepared <- df_prepared |> select(-state, -year)
```

6.2.2 Data split

Finally, we split the data 70:30 in preparation for model development.

```
set.seed(144)
train_indices <- createDataPartition(df_prepared$pain_relievers_misuse, p = 0.7, list = FALSE)
oud_train_split <- df_prepared[train_indices, ]
oud_test_split <- df_prepared[-train_indices, ]
```

6.3 Results of Assumption checks (Nana)

6.4 Modeling

Before building the models, we define the 10-fold Cross-Validation logic that we will use for model selection and evaluation.

```
# Define 10-fold Cross-Validation logic
# Ref: AMAR Ch 7.3 (Predictive Ability)
train_control <- trainControl(method = "cv", number = 10)

# List to contain modeling objects and cross-validation results
model_results_list <- list()
```

Our First model is the Full Multiple Linear Regression model, which includes all predictors. This model serves as a baseline for comparison with more parsimonious models.

```
# 1. Full Multiple Linear Regression
model_full <- train(
  pain_relievers_misuse ~ .,
  data = oud_train_split,
  method = "lm",
  trControl = train_control
)

model_full
```

Linear Regression

144 samples

20 predictor

No pre-processing

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 130, 129, 129, 132, 130, 130, ...

Resampling results:

RMSE	Rsquared	MAE
0.008134775	0.2633035	0.006542457

Tuning parameter 'intercept' was held constant at a value of TRUE

```
model_results_list$Full_MLR <- model_full
```

Secondly, we will train our Stepwise Selection model, which iteratively adds or removes predictors based on AIC to find a more parsimonious model.

```
# 2. Stepwise Selection based on AIC
model_step <- train(
  pain_relievers_misuse ~ .,
  data = oud_train_split,
  method = "lmStepAIC",
  trControl = train_control,
  trace = FALSE
)

model_step
```

Linear Regression with Stepwise Selection

144 samples
20 predictor

No pre-processing

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 130, 130, 131, 130, 129, 129, ...

Resampling results:

RMSE	Rsquared	MAE
0.007938368	0.2375276	0.006570298

```
model_results_list$Stepwise_AIC <- model_step
```

Next, we train our LASSO regression model, which applies a penalty to shrink less important coefficients to zero, effectively performing variable selection. We implement a grid search to find the optimal value of the regularization parameter λ that minimizes the cross-validated error.


```
# 3. LASSO Regression with Hyperparameter Tuning
lasso_grid <- expand.grid(fraction = seq(0.1, 1, length = 10))

model_lasso <- train(
  pain_relievers_misuse ~ .,
  data = oud_train_split,
  method = "lasso",
  trControl = train_control,
  tuneGrid = lasso_grid
)

model_lasso
```

The lasso

144 samples

20 predictor

No pre-processing

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 130, 130, 129, 129, 129, 132, ...

Resampling results across tuning parameters:

fraction	RMSE	Rsquared	MAE
0.1	0.007728257	0.2588801	0.006212928
0.2	0.007867598	0.2504116	0.006315973
0.3	0.007929956	0.2487620	0.006404565
0.4	0.007921102	0.2525857	0.006410815
0.5	0.007916573	0.2561573	0.006407290
0.6	0.007906253	0.2611757	0.006384530
0.7	0.007896825	0.2667679	0.006364264
0.8	0.007910281	0.2683819	0.006379110
0.9	0.007929799	0.2698583	0.006391412
1.0	0.007964705	0.2702747	0.006409759

RMSE was used to select the optimal model using the smallest value.

The final value used for the model was fraction = 0.1.

```
model_results_list$LASSO <- model_lasso
```

Lastly, we will compare the performance of these three models based on their cross-validated RMSE and select the best performing model for final evaluation on the test set.

```
# Collecting CV results for comparison
cv_summary <- resamples(list(
  Full = model_results_list$Full_MLR,
  Stepwise = model_results_list$Stepwise_AIC,
  LASSO = model_results_list$LASSO
))

summary(cv_summary)
```

Call:

```
summary.resamples(object = cv_summary)
```

Models: Full, Stepwise, LASSO

Number of resamples: 10

MAE

	Min.	1st Qu.	Median	Mean	3rd Qu.
Full	0.005392517	0.006097335	0.006347721	0.006542457	0.006708942
Stepwise	0.004522811	0.005781265	0.006854365	0.006570298	0.007492335
LASSO	0.004355174	0.006085059	0.006320877	0.006212928	0.006711527

	Max.	NA's
Full	0.008036912	0
Stepwise	0.007695908	0
LASSO	0.007588355	0

RMSE

	Min.	1st Qu.	Median	Mean	3rd Qu.
Full	0.006458636	0.007418801	0.008002010	0.008134775	0.008819530
Stepwise	0.005656253	0.006872333	0.008239586	0.007938368	0.009044640
LASSO	0.005949474	0.007496975	0.007688121	0.007728257	0.008197995

	Max.	NA's
Full	0.010042840	0
Stepwise	0.009501905	0
LASSO	0.009275562	0

Rsquared

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
Full	0.01311226	0.1249150	0.2188552	0.2633035	0.3992077	0.5455291	0
Stepwise	0.01022618	0.1370386	0.2399172	0.2375276	0.3779030	0.4149258	0
LASSO	0.07537612	0.1787170	0.2877682	0.2588801	0.3219396	0.4198144	0

6.5 Results of model selection (Said)

7 Discussion

This study has several limitations. Namely, our data are at the state level, so we cannot make inferences about individual-level risk factors for OUD. In addition, our data are cross-sectional, so we cannot make causal inferences about the relationships between predictors and OUD. Finally, our data only include a limited number of predictors, so there may be other important factors that we did not include in our models.

8 Conclusions (Naomi)

We found that...

Future research could...

Conclude your findings, limitations, and suggest areas for future work.

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